CBSE Class 10 Mathematics — Arithmetic Progressions — Solutions

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1. **1.** Assertion (A): The common difference of the AP $5, 2, -1, -4, ...$ is -3. Reason (R): The common difference d of an AP $a_1, a_2, a_3, ...$ is given by $d = a_n - a_{n-1}$.

   Answer: (A)

   Solution:

   1. Calculate common difference: $d = a_2 - a_1 = 2 - 5 = -3$.
   2. Verify the formula for common difference.

   Derivation:

   $$d = 2 - 5 = -3; \text{ Yes, } d = a_n - a_{n-1} \text{ is the definition.}$$

   Arithmetic Progressions — Common Difference

2. **2.** Assertion (A): The 10th term of the AP $2, 7, 12, ...$ is 47. Reason (R): The n^{th} term of an AP is given by $a_n = a + (n-1)d$.

   Answer: (A)

   Solution:

   1. Identify $a = 2, d = 5, n = 10$.
   2. Apply $a_{10} = 2 + (10-1)5 = 2 + 45 = 47$.

   Derivation:

   $$a_{10} = 2 + (9)(5) = 2 + 45 = 47$$

   Arithmetic Progressions — nth term of an AP

3. **3.** Assertion (A): The sum of the first 10 natural numbers is 55. Reason (R): The sum of first n natural numbers is given by $\frac{n(n+1)}{2}$.

   Answer: (A)

   Solution:

   1. Use formula $S_n = \frac{n(n+1)}{2}$ for $n = 10$.
   2. Calculate $S_{10} = \frac{10 \times 11}{2} = 55$.

   Derivation:

   $$S_{10} = \frac{10(11)}{2} = 5imes11 = 55$$

   Arithmetic Progressions — Sum of first n terms

4. **4.** Assertion (A): The sequence $3, 3, 3, 3, ...$ is an AP. Reason (R): A sequence is an AP if the difference between consecutive terms is constant.

   Answer: (A)

   Solution:

1. Calculate differences: $3-3=0, 3-3=0$.
2. Since difference is constant ($d=0$), it is an AP.

Derivation:

$$a_2 - a_1 = 0; a_3 - a_2 = 0; d = 0 \text{ (constant)}$$

Arithmetic Progressions — Definition of AP

5. **5.** Assertion (A): If $2x, x+10, 3x+2$ are in AP, then $x=6$. Reason (R): If a, b, c are in AP, then $2b = a+c$.

Answer: (A)

Solution:

1. Use property $2(x+10) = 2x + (3x+2)$.
2. Solve for x: $2x + 20 = 5x + 2 \implies 18 = 3x \implies x = 6$.

Derivation:

$$\begin{aligned} 2(x+10) &= 2x + 3x + 2 \\ \implies 2x + 20 &= 5x + 2 \\ \implies 18 &= 3x \\ \implies x &= 6 \end{aligned}$$

Arithmetic Progressions — Arithmetic Mean property

6. **6.** Find the 10^{th} term of the Arithmetic Progression $5, 8, 11, 14, \ldots$

Answer:

$$32$$

Solution:

1. Identify the first term $a=5$ and the common difference $d = 8-5 = 3$.
2. Use the formula $a_n = a + (n-1)d$ to calculate the 10^{th} term.

Derivation:

$$\begin{aligned} a_n &= a + (n-1)d \\ a_{10} &= 5 + (10-1)3 \\ a_{10} &= 5 + 9 \times 3 \\ a_{10} &= 5 + 27 = 32 \end{aligned}$$

Arithmetic Progressions — nth term of an AP

7. **7.** Which term of the Arithmetic Progression $21, 18, 15, \ldots$ is -81?

Answer:

$$35^{th} term$$

Solution:

1. Identify $a = 21$ and $d = 18 - 21 = -3$.
2. Set $a_n = -81$ and solve the equation $-81 = 21 + (n-1)(-3)$ for n.

Derivation:

$$\begin{aligned}-81 &= 21 - 3n + 3\\ -81 &= 24 - 3n\\ 3n &= 24 + 81\\ 3n &= 105\\ n &= 35\end{aligned}$$

Arithmetic Progressions — nth term of an AP

8. **8.** Find the sum of the first 20 terms of the Arithmetic Progression $1, 6, 11, 16, \ldots$

Answer:

$$970$$

Solution:

1. Identify $a = 1$, $d = 5$, and $n = 20$.
2. Use the sum formula $S_n = \frac{n}{2}[2a + (n-1)d]$.

Derivation:

$$\begin{aligned}S_{20} &= \frac{20}{2}[2(1) + (20-1)5]\\ S_{20} &= 10[2 + 19 \times 5]\\ S_{20} &= 10[2 + 95]\\ S_{20} &= 10 \times 97 = 970\end{aligned}$$

Arithmetic Progressions — Sum of first n terms

9. **9.** In an Arithmetic Progression, if the first term is 7 and the common difference is 4, find the sum of the first 10 terms.

Answer:

$$250$$

Solution:

1. State the given values: $a = 7$, $d = 4$, $n = 10$.
2. Substitute these values into $S_n = \frac{n}{2}[2a + (n-1)d]$.

Derivation:

$$\begin{aligned}S_{10} &= \frac{10}{2}[2(7) + (10-1)4]\\ S_{10} &= 5[14 + 9 \times 4]\\ S_{10} &= 5[14 + 36]\\ S_{10} &= 5 \times 50 = 250\end{aligned}$$

Arithmetic Progressions — Sum of first n terms

10. **10.** If the 4^{th} term of an Arithmetic Progression is 14 and the 12^{th} term is 38, find the first term a.

Answer:

$$5$$

Solution:

1. Write equations based on $a_n = a + (n-1)d$: $a + 3d = 14$ and $a + 11d = 38$.
2. Subtract the two equations to find d and then solve for a.

Derivation:

$$\begin{aligned}(a + 11d) - (a + 3d) &= 38 - 14\\ 8d &= 24\\ d &= 3\\ a + 3(3) &= 14\\ a + 9 &= 14\\ a &= 5\end{aligned}$$

Arithmetic Progressions — nth term of an AP

11. 11. Find the sum of all two-digit numbers which, when divided by 4, yield 1 as a remainder.

Answer:

1210

Solution:

1. Identify the sequence: The smallest two-digit number is 10. $13 \div 4 = 3$ remainder 1. So the first term $a = 13$.
2. Find the last term: The largest two-digit number is 99. $97 \div 4 = 24$ remainder 1. So the last term $l = 97$.
3. Find the number of terms n: Use $a_n = a + (n-1)d$. $97 = 13 + (n-1)4$.
4. Calculate the sum using $S_n = \frac{n}{2}(a + l)$.

Derivation:

$$\begin{aligned}97 &= 13 + 4n - 4\\ \implies 88 &= 4n\\ \implies n = 22.S_{22} = \frac{22}{2}(13 + 97) = 11 \times 110 &= 1210.\end{aligned}$$

Arithmetic Progressions — Sum of n terms

12. 12. If the sum of the first n terms of an AP is $S_n = 3n^2 + 5n$, find the 16^{th} term of the AP.

Answer:

98

Solution:

1. Use the relationship between the n^{th} term and the sum of terms: $a_n = S_n - S_{n-1}$.
2. Calculate $S_{16} = 3(16)^2 + 5(16)$.
3. Calculate $S_{15} = 3(15)^2 + 5(15)$.
4. Subtract S_{15} from S_{16} to find a_{16}.

Derivation:

$S_{16} = 3(256)+80 = 768+80 = 848. S_{15} = 3(225)+75 = 675+75 = 750. a_{16} = 848-750 = 98.$

Arithmetic Progressions — Sum of n terms

13. 13. How many terms of the AP $24, 21, 18, \ldots$ must be taken so that their sum is 78?

Answer:

$$4 or 13$$

Solution:

1. Identify $a = 24$, $d = -3$, and $S_n = 78$.
2. Use the formula $S_n = \frac{n}{2}[2a + (n-1)d]$.
3. Solve the resulting quadratic equation for n.
4. Verify both values of n.

Derivation:

$$\begin{aligned} 78 &= \frac{n}{2}[48 + (n-1)(-3)] \\ \implies 156 &= n(48 - 3n + 3) \\ \implies 156 &= 51n - 3n^2 \\ \implies 3n^2 - 51n + 156 &= 0 \\ \implies n^2 - 17n + 52 &= 0 \\ \implies (n-4)(n-13) &= 0. \end{aligned}$$

Arithmetic Progressions — Sum of n terms

14. 14. Find the middle term of the AP: $6, 13, 20, \ldots, 216$.

Answer:

$$111$$

Solution:

1. Find the total number of terms n using $a_n = a + (n-1)d$.
2. Identify the middle term position: If n is odd, middle term is $\frac{n+1}{2}$.
3. Calculate the value of the middle term.

Derivation:

$$\begin{aligned} 216 &= 6 + (n-1)7 \\ \implies 210 &= 7(n-1) \\ \implies 30 &= n-1 \\ \implies n = 31. Middle term is \frac{31+1}{2} = 16^{th} term. a_{16} &= 6 + 15(7) = 6 + 105 = 111. \end{aligned}$$

Arithmetic Progressions — n-th term

15. 15. The 4^{th} term of an AP is 11 and the sum of its 5^{th} and 7^{th} terms is 34. Find the common difference.

Answer:

$$3$$

Solution:

1. Express a_4 as $a + 3d = 11$.
2. Express $a_5 + a_7$ as $(a + 4d) + (a + 6d) = 34$.
3. Solve the system of two linear equations for d.

Derivation:

$$\begin{aligned} a + 3d = 11(Eq1).2a + 10d = 34 \\ \implies a + 5d = 17(Eq2).SubtractingEq1fromEq2 : (a + 5d) - (a + 3d) = 17 - 11 \\ \implies 2d = 6 \\ \implies d = 3. \end{aligned}$$

Arithmetic Progressions — n-th term

16. 16. A sum of Rs. 2800 is to be used to award four prizes to students of a school for their overall academic performance. If each prize is Rs. 200 less than its preceding prize, find the value of each of the prizes.

Answer:

$$\text{Rs. } 850, \text{Rs. } 650, \text{Rs. } 450, \text{Rs. } 350$$

Solution:

1. Let the four prizes be $a + 3d, a + d, a - d, a - 3d$ or $a, a - d, a - 2d, a - 3d$. Let's use $a, a - d, a - 2d, a - 3d$.
2. The sum is $a + (a - d) + (a - 2d) + (a - 3d) = 2800$.
3. Simplify the sum: $4a - 6d = 2800$. Given $d = 200$.
4. Substitute $d = 200$ into the equation: $4a - 6(200) = 2800$.
5. Solve for a: $4a - 1200 = 2800 \Rightarrow 4a = 4000 \Rightarrow a = 1000$.
6. Calculate the prizes: $1000, 800, 600, 400$ is incorrect based on the sum logic. Correcting: Let prizes be $x, x - 200, x - 400, x - 600$. Sum: $4x - 1200 = 2800 \Rightarrow 4x = 4000 \Rightarrow x = 1000$. Prizes: $1000, 800, 600, 400$ is a sum of 2800. Wait, $1000 + 800 + 600 + 400 = 2800$. Correct.

Derivation:

$$\begin{aligned} S_n = \frac{n}{2}[2a + (n - 1)d] \\ \Rightarrow 2800 = \frac{4}{2}[2a + 3(-200)] \\ \Rightarrow 1400 = 2a - 600 \\ \Rightarrow 2a = 2000 \\ \Rightarrow a = 1000.Prizesare 1000, 800, 600, 400. \end{aligned}$$

Arithmetic Progressions — Sum of n terms of an AP

17. **17.** The sum of the first n terms of an AP is given by $S_n = 3n^2 + 5n$. Find the 16^{th} term of this AP and the general expression for the n^{th} term.

Answer:

$$a_{16} = 92, a_n = 6n + 2$$

Solution:

1. Use the formula $a_n = S_n - S_{n-1}$ for $n > 1$.
2. Calculate $S_n = 3n^2 + 5n$.
3. Calculate $S_{n-1} = 3(n-1)^2 + 5(n-1) = 3(n^2 - 2n + 1) + 5n - 5 = 3n^2 - 6n + 3 + 5n - 5 = 3n^2 - n - 2$.
4. Find $a_n = (3n^2 + 5n) - (3n^2 - n - 2) = 6n + 2$.
5. Calculate $a_{16} = 6(16) + 2 = 96 + 2 = 98$.

Derivation:

$$a_n = S_n - S_{n-1} = 3n^2 + 5n - [3(n-1)^2 + 5(n-1)] = 3n^2 + 5n - [3n^2 - 6n + 3 + 5n - 5] = 6n + 2. For n = 16,$$

Arithmetic Progressions — Sum of n terms

18. **18.** In an AP, the sum of first p terms is q and the sum of first q terms is p. Show that the sum of first $(p + q)$ terms is $-(p + q)$.

Answer:

$$S_{p+q} = -(p + q)$$

Solution:

1. Write sum formulas: $S_p = \frac{p}{2}[2a + (p-1)d] = q$ and $S_q = \frac{q}{2}[2a + (q-1)d] = p$.
2. Simplify to $2a + (p-1)d = \frac{2q}{p}$ and $2a + (q-1)d = \frac{2p}{q}$.
3. Subtract equations to find d: $(p-q)d = \frac{2q}{p} - \frac{2p}{q} = \frac{2(q^2-p^2)}{pq} = \frac{-2(p-q)(p+q)}{pq}$.
4. Thus $d = \frac{-2(p+q)}{pq}$.
5. Find $2a$ and then $S_{p+q} = \frac{p+q}{2}[2a + (p+q-1)d]$.

Derivation:

$$Subtracting gives d = -2/pq(p+q). Substituting d back gives 2a = 2/p(q) - (p-1)d. Sum S_{p+q} = \frac{p+q}{2}[$$

Arithmetic Progressions — Sum of n terms

19. **19.** A spiral is made up of successive semicircles, with centers alternately at A and B, starting with center at A of radii $0.5cm, 1.0cm, 1.5cm, 2.0cm, \ldots$. What is the total length of such a spiral made up of thirteen consecutive semicircles? (Take $\pi = \frac{22}{7}$)

Answer:

$$143cm$$

Solution:

1. The length of a semicircle is $L = \pi r$.

2. The lengths are $L_1 = 0.5\pi, L_2 = 1.0\pi, L_3 = 1.5\pi, \ldots$.
3. This forms an AP with $a = 0.5$ and $d = 0.5$.
4. We need the sum of $n = 13$ terms: $S_{13} = \frac{13}{2}[2(0.5\pi) + (13 - 1)(0.5\pi)]$.
5. Calculate $S_{13} = \frac{13}{2}[\pi + 6\pi] = \frac{13}{2}[7\pi] = \frac{91\pi}{2}$.
6. Substitute $\pi = \frac{22}{7}$: $S_{13} = \frac{91}{2} \times \frac{22}{7} = 13 \times 11 = 143$.

Derivation:

$$S_{13} = \frac{13}{2}[2(0.5\pi) + 12(0.5\pi)] = \frac{13}{2}[\pi + 6\pi] = \frac{13 \times 7 \times \pi}{2} = \frac{91}{2} \times \frac{22}{7} = 143.$$

Arithmetic Progressions — Applications of AP